PhysioPose: Real-Time Squat Form Classification from a Standard Webcam

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1. Motivation & Problem Statement

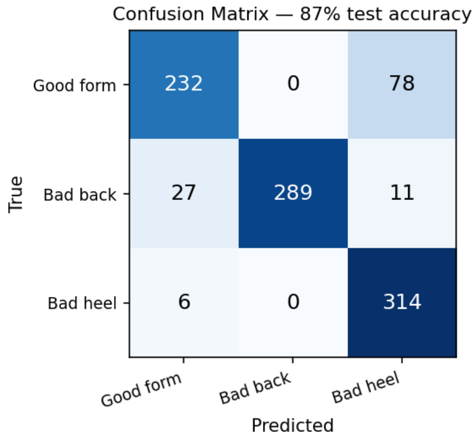
Home-based physiotherapy and strength rehabilitation depend on patients performing exercises with correct form, yet most rehabilitation happens unsupervised, where poor technique reduces effectiveness and can cause re-injury. The squat is one of the most prescribed lower-limb rehabilitation and strengthening movements, but common faults—excessive forward/back lean and lifting the heels—are easy to make and hard for an untrained person to self-detect. Clinical motion-capture systems and wearable sensors can provide feedback, but they are expensive, require setup, and are inaccessible for everyday at-home use. There is a clear gap for a low-cost tool that gives immediate form feedback using only hardware the user already owns: a webcam. Our engineering goal is to build a system that classifies squat form in real time from a single RGB webcam and tells the user whether their form is good or which specific fault they are making. We build on Google's MediaPipe Pose, which made robust on-device 2D human-pose estimation widely available [1], and on prior work showing joint-angle features are strong predictors of exercise quality [3]. The result is intended to make rehabilitation feedback more accessible and affordable without any specialized hardware.

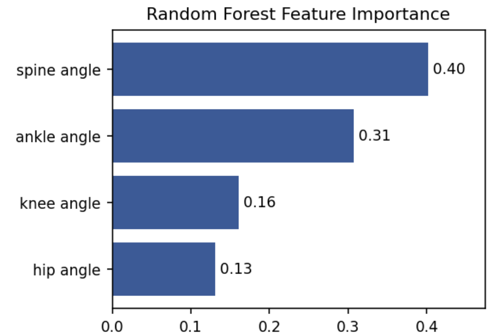
2. Approach

PhysioPose is a real-time pipeline: each webcam frame is passed to MediaPipe Pose, which returns 33 body landmarks; we then convert the relevant landmarks into a small set of biomechanical features and classify the current posture. Rather than feeding raw pixel coordinates to the model, we compute four *resolution-invariant joint angles*—knee, hip, ankle, and spine (torso) angle—using vector geometry between landmark triples. These angles capture exactly the cues a physiotherapist watches: knee/hip angle for squat depth, spine angle for torso lean (the "bad back" fault), and ankle angle for heel lift (the "bad heel" fault). The features are fed to a Random Forest classifier (scikit-learn, 200 trees, balanced class weights) that outputs one of three labels—**Good form**, **Bad back**, or **Bad heel**—with a confidence score that is overlaid live on the video. We trained on a public side-view squat image dataset (Zenodo [2], ∼3,800 images), running every image through the *same* MediaPipe + angle-extraction code used at inference so that training and live features are produced identically. The primary tools are Python 3.10, OpenCV (capture/overlay), MediaPipe Pose (landmarks), and scikit-learn/pandas (features and model). Key design decisions were (i) using angle features instead of pixel coordinates for camera- and resolution-independence, and (ii) extracting from the single camera-facing leg so that side-view occlusion of the far leg does not discard frames. Scope and assumptions: the user stands side-on, full body in frame, one person, and we classify a fixed set of three squat faults rather than arbitrary exercises.

3. Key Results

On a held-out test split of the Zenodo benchmark, the Random Forest reaches **87% accuracy** across the three classes (left). Performance is strongest on **Bad back**, with **1.00 precision** and 0.88 recall—torso lean is a clean, well-separated signal, which the feature-importance chart (right) confirms: the spine angle is the single most informative feature, followed by the ankle angle that drives heel detection. The dominant error is Good-vs-Bad heel confusion (78 of 310 "Good" test frames are read as "Bad heel"), because the difference between a slightly raised heel and a flat foot is small and noisy in 2D. The most important practical finding was a *domain-shift* effect: a model trained on the high-resolution benchmark photos degraded sharply on our own 480p webcam, where joint angles read systematically 15–20° higher. Retraining on a small set of self-recorded webcam squats brought the system in-domain and made live prediction usable, though with limited recorded data (∼1,000 frames) in-domain accuracy was lower (∼63%), again with Good/Bad-heel as the hard pair. This matched our expectation that feature *parity* between training and inference matters more than raw dataset size, and it cleanly separates the two faults that are geometrically distinct (lean, heel) from the harder near-threshold cases.





4. Challenges & Lessons Learned

Our first dataset was synthetic—“bad” squats were generated by adding noise to correct ones—and the resulting model collapsed to predicting a single class regardless of input; it could not even recover its own training-class means. The lesson was to validate a model against its own data distribution before trusting accuracy numbers, and that realistic data beats synthetic perturbations. The second major issue was a train/inference mismatch: features computed in raw pixels (e.g. hip depth) were $\sim 7\times$ larger on the 3456 px training images than on the 480p webcam, so the model keyed on splits that could never fire live—we fixed this by switching to scale-invariant angle features. Side-view occlusion of the far leg initially discarded most frames, which we solved by extracting features from the single visible leg and relaxing the landmark-visibility threshold. Finally, the photo-to-webcam domain gap forced us to record our own calibration data on the target camera. The overarching lesson: in a perception pipeline, guaranteeing that training and live features come from the identical code path is as important as the model itself.

5. Individual Contributions

Member	Primary Contributions	Approx. %
Kanishk Kaushal	Feature extraction, ML training/evaluation, dataset pipeline, domain-shift debugging, live integration, and documentation; plus shared foundational work below	75%
Kaushal Jha	Repository setup, dependency configuration, webcam capture pipeline, pose-estimation integration, and dataset sourcing (jointly with Kanishk)	25%

6. Repository & References

GitHub Repository: `https://github.com/Kanishk-Kaushal/PhysioPose`
Project Tracker: `https://github.com/users/Kanishk-Kaushal/projects/3/views/1`
Ensure the repo is public, or add the instructor as a collaborator.

Key References:

[1] C. Lugaresi et al., “MediaPipe: A Framework for Building Perception Pipelines,” Google Research, 2019. Real-time on-device 2D pose estimation used for landmark detection.

[2] C. Teng, “Squat Posture Image Dataset (Good / Bad Back / Bad Heel),” Zenodo, 2025, record 17558630. Side-view labeled images used for training and benchmarking.

[3] D. R. Beddiar et al., “Vision-based human activity recognition for rehabilitation,” (survey context for pose-based exercise assessment).